

Supplementary Results: Naive Baseline Comparison

Hypoglycaemia Event Detection – T1DATA Cohort (36 Patients)

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Overview

This document reports the performance of two naive rule-based detectors alongside the trained event classifiers. The naive baselines require no learning and serve as a lower bound: a trained model that does not outperform them has not extracted any useful information from the data.

All metrics are cohort means over 36 patients, evaluated on the held-out test split.

Naive Baseline Definitions

Naive Last Value. If the most recent glucose reading in the lookback window is below 70 mg/dL, an event is predicted. This rule fires only when the patient is already in hypoglycaemia. It is highly precise but misses the large majority of events because it cannot anticipate future drops.

Naive Mean. If the average glucose over the full 2-hour lookback window is below 70 mg/dL, an event is predicted. Because the average must fall below the threshold, this rule fires very rarely and misses almost everything.

Task A: Will a hypoglycaemia event occur anywhere in the next 60 minutes?

Table 1: Task A results. F2 weights recall twice as heavily as precision.

Model	Precision	Recall	F1	F2
Naive Last Value	0.857	0.326	0.473	0.373
Naive Mean	0.396	0.041	0.074	0.050
LSTM Event Classifier	0.237	0.693	0.334	0.463
PatchTST Event Classifier	0.054	0.615	0.095	0.178

The Naive Last Value achieves a higher F1 (0.473) than the LSTM classifier (0.334) because it operates at a very different precision–recall trade-off: it fires only when glucose is already below 70 mg/dL, which makes it very precise but nearly useless for early warning. The LSTM classifier has a substantially higher recall (0.693 vs. 0.326), meaning it catches more than twice as many events at an earlier point in time. On F2, which weights recall more heavily, the LSTM classifier (0.463) outperforms the naive rule (0.373). The PatchTST classifier does not outperform the Naive Last Value on any metric.

Task B: Will a hypoglycaemia event occur at approximately 60 minutes (tolerance ± 15 min)?

Table 2: Task B results. The positive label is 1 if the minimum glucose in the window [45, 75] min ahead is below 70 mg/dL.

Model	Precision	Recall	F1	F2
Naive Last Value	0.216	0.113	0.149	0.125
Naive Mean	0.115	0.016	0.029	0.020
LSTM Point-Event Classifier	0.088	0.525	0.143	0.237
PatchTST Point-Event Classifier	0.036	0.537	0.064	0.124

Task B is a harder and more specific question. The LSTM point-event classifier barely exceeds the Naive Last Value on F1 (0.143 vs. 0.149) and has substantially lower precision (0.088 vs. 0.216). On F2, the LSTM (0.237) outperforms the naive rule (0.125) due to its higher recall. The PatchTST point-event classifier does not outperform the Naive Last Value on any metric. These results indicate that predicting the exact 60-minute horizon is a considerably harder problem than predicting whether any event will occur in the next hour.

Note on Inclusion in Thesis

The naive baselines are reported here for completeness and supervisor reference. Following discussion with the supervisor, they are not included in the main thesis, as the focus of the ablation study is on the comparison between forecasting-derived detectors and trained binary classifiers rather than on rule-based baselines.